



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSIC ACADEMY

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Course Code: SBT-DF203

Course Title: Basic Computer Skills for Digital Forensics

Lab: Lab 9 — WEP40 Wireless Packet Decryption and Aircrack Forensics

Operating System: Kali Linux

Laboratory Workspace: SBT-DF203_Lab9_2025_FWSD_11470_Peter_Oloo

Submission Date: 25 Sept 2026

Create Lab 9 workspace

```
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203_Lab9_Peter_Oloo/{evidence,working,pcap,hashes,screenshots,outputs,recovered,logs,report}

cd ~/SBT-DF203_Lab9_Peter_Oloo

pwd
/home/kali/SBT-DF203_Lab9_Peter_Oloo
find . -maxdepth 1 -type d -printf '%f\n' | sort
./
evidence/
hashes/
logs/
outputs/
pcap/
recovered/
report/
screenshots/
working/

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$
```

Supplied historical evidence

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ cd ~/SBT-DF203_Lab9_Peter_Oloo/evidence

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ wget -O file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-09-25 02:13:49-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'file.xz'

file.xz                                     100%[=====] 12.24M 3.02MB/s in 4.4s

2026-09-25 02:13:58 (2.81 MB/s) - 'file.xz' saved [12837660/12837660]

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ ls -lh file.xz
file file.xz
-rw-rw-r-- 1 kali kali 13M Sep 25 02:13 file.xz
file.xz: XZ compressed data, checksum CRC64

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ sha256sum file.xz | tee ../hashes/file.xz.sha256
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ cat ../hashes/file.xz.sha256
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$

```

Verify the compressed evidence

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ xz -t file.xz
echo $?
0

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ file file.xz
file.xz: XZ compressed data, checksum CRC64

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$

```

Make the working copy

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/evidence]
$ cd ~/SBT-DF203_Lab9_Peter_Oloo

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ cp --preserve=all evidence/file.xz working/file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ sha256sum evidence/file.xz working/file.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ sha256sum evidence/file.xz working/file.xz | tee hashes/working_copy_verification.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ cat hashes/working_copy_verification.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$

```

Decompress the working copied

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ cd ~/SBT-DF203_Lab9_Peter_Oloo/working

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ xz -dk file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ ls -lh
file file
total 26M
-rw-rw-r-- 1 kali kali 14M Sep 25 02:13 file
-rw-rw-r-- 1 kali kali 13M Sep 25 02:13 file.xz
file: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$

```

Identify the capture

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ capinfos file
File name: file
File type: Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 45 k
File size: 14 MB
Data size: 13 MB
Capture duration: 272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time: 2015-03-06 02:56:27.334320
Data byte rate: 49 kBps
Data bit rate: 395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1: cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 45169

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ file file
file: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ ls -lh file
-rw-rw-r-- 1 kali kali 14M Sep 25 02:13 file

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$

```

Copy the capture into the PCAP evidence area

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ cp --preserve=all ~/SBT-DF203_Lab9_Peter_Oloo/working/file \
~/SBT-DF203_Lab9_Peter_Oloo/pcap/wep40_original_working.pcap

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$ sha256sum ~/SBT-DF203_Lab9_Peter_Oloo/pcap/wep40_original_working.pcap \
| tee ~/SBT-DF203_Lab9_Peter_Oloo/pcap/wep40_working_sha256.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d /home/kali/SBT-DF203_Lab9_Peter_Oloo/pcap/wep40_original_working.pcap

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo/working]
$

```

Examine the 802.11 traffic


```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ tshark -r pcap/wep40_original_working.pcap \
-Y 'wlan' \
-T fields \
-e frame.number \
-e frame.time_relative \
-e wlan.fc.type \
-e wlan.fc.subtype \
-e wlan.sa \
-e wlan.da \
-e wlan.bssid \
| head -50
1      0.000000000    1      11
2      0.000001000    1      11
3      -0.000001000   1      11
4      0.000000000    1      11
5      0.001024000    1      11
6      0.001537000    1      11
7      0.001536000    1      11
8      0.001535000    1      11
9      0.001537000    1      11
10     0.001535000    1      11
11     0.001538000    1      11
12     0.001537000    1      11
13     0.001537000    1      11
14     0.001537000    1      11
15     0.001538000    1      11

```

Identify WEP-protected traffic

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ tshark -r pcap/wep40_original_working.pcap \
-Y 'wlan.wep' \
-T fields \
-e frame.number \
-e wlan.bssid \
-e wlan.sa \
-e wlan.da \
-e wlan.wep.iv \
-e wlan.wep.key
tshark: "wlan.wep" is not a valid protocol or protocol field.
wlan.wep
^~~~~~

```

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ tshark -r pcap/wep40_original_working.pcap \
-Y 'wlan.wep' \
-T fields \
-e frame.number \
-e wlan.bssid \
-e wlan.sa \
-e wlan.da \
-e wlan.wep.iv \
-e wlan.wep.key \
> outputs/wep_iv_inventory.txt
tshark: "wlan.wep" is not a valid protocol or protocol field.
wlan.wep
^~~~~~

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ head -30 outputs/wep_iv_inventory.txt

```

Aircrack-ng

```
(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ which aircrack-ng
aircrack-ng --help | head
/usr/bin/aircrack-ng

Aircrack-ng 1.7 - (C) 2006-2022 Thomas d'Otreppe
https://www.aircrack-ng.org

usage: aircrack-ng [options] <input file(s)>

Common options:

-a <amode> : force attack mode (1/WEP, 2/WPA-PSK)
-e <essid> : target selection: network identifier
```

```
(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ aircrack-ng pcap/wep40_original_working.pcap
Reading packets, please wait ...
Opening pcap/wep40_original_working.pcap
Read 45169 packets.

# BSSID          ESSID          Encryption
1 00:26:66:55:97:D6 cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait ...
Opening pcap/wep40_original_working.pcap
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0  0/ 1  A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688)
1  0/ 1  3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) BD(19456) 80(19200) 85(19200) 9D(19200) 13(18944)
2  0/ 1  F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688)
3  1/ 10 F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$
```

Hash the decrypted capture

```

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ airdecap-ng -w A4:3D:F6:F3:74 pcap/wep40_original_working.pcap
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ ls -lh pcap/
total 26M
-rw-rw-r-- 1 kali kali 13M Sep 25 03:20 wep40_original_working-dec.pcap
-rw-rw-r-- 1 kali kali 14M Sep 25 02:13 wep40_original_working.pcap

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ find . -maxdepth 2 -type f -printf '%p\n' | sort
./evidence/file.xz
./hashes/file.xz.sha256
./hashes/wep40_working_sha256.txt
./hashes/working_copy_verification.txt
./outputs/protocol_hierarchy.txt
./outputs/wep_iv_inventory.txt
./pcap/wep40_original_working-dec.pcap
./pcap/wep40_original_working.pcap
./working/file
./working/file.xz

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$ file pcap/*
pcap/wep40_original_working-dec.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 65535)
pcap/wep40_original_working.pcap:    pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203_Lab9_Peter_Oloo]
$

```

Summary report

In this lab, we conducted a forensic analysis of a given historical IEEE 802.11 wireless packet capture secured with WEP40. The study was done offline based on the historical evidence provided.

The investigation began by preserving the provided compressed evidence file, file.xz, and computing the SHA-256 hash. Evidence was tested successfully with xz -t and a working copy was produced. The hash of the working copy was SHA-256 which matched the original evidence.

The evidence was uncompressed and determined to be an IEEE 802.11 PCAP with 45,169 packets. The capture has 15,713 WLAN data frames and took about 272.97 seconds.

Aircrack-ng 1.7 identified one WEP-protected wireless network:

- **ESSID:** cgnetwork
- **BSSID:** 00:26:66:55:97:D6
- **Encryption:** WEP
- **Captured WEP IVs:** 15,477

Aircrack-ng recovered the WEP40 key:

A4:3D:F6:F3:74

and reported:

Decrypted correctly: 100%

The recovered key was then given to Airdecap-ng. Airdecap-ng processed all 45,169 packets successfully and decrypted all 15,477 WEP data packets. It reported that there were zero corrupted WEP packets.

The capture was produced as decrypted:

pcap/wep40_original_working-dec.pcap

The decrypted capture was identified as Ethernet encapsulated PCAP and the original working capture was identified as IEEE 802.11 PCAP.

The laboratory has successfully demonstrated evidence preservation, WEP40 key recovery, and offline WEP packet decryption. Therefore, those results are not fabricated in this report because there is no completed post-decryption endpoint/conversation or recovered-object analysis in the present evidence record.

Investigation Environment

The workspace contained the following directories:

```
evidence/  
working/  
pcap/  
hashes/  
screenshots/  
outputs/  
recovered/  
logs/  
report/
```

Capture Metadata

File type Wireshark/tcpdump PCAP

Encapsulation IEEE 802.11 Wireless LAN

Timestamp precision Microseconds

Number of packets 45,169

File size 14 MB

Data size 13 MB

Capture duration 272.969795 seconds
Earliest packet 2015-03-06 02:51:54.364525
Latest packet 2015-03-06 02:56:27.334320
Capture length 65,535 bytes
Interfaces 1
Interface encapsulation IEEE 802.11 Wireless LAN
Strict time order False

Initial Airdecap-ng Error

The first Airdecap-ng attempt used:

```
airdecap-ng -w KEY pcap/wep40_original_working.pcap
```

The command failed with:

```
Invalid WEP key length. [5,13,16,29,61]
```

The error resulted from using the placeholder `KEY` rather than the actual recovered key.

The actual recovered key was:

```
A4:3D:F6:F3:74
```

Successful Airdecap-ng Decryption

Total number of stations seen 10 Total number of packets read 45169 Total number of WEP data packets 15477 Total number of WPA data packets 0 Number of plaintext data packets 0 Number of decrypted WEP packets 15477 Number of corrupted WEP packets 0 Number of decrypted WPA packets 0 Number of bad TKIP (WPA) packets 0 Number of bad CCMP (WPA) packets 0

Conclusion

The WEP40 wireless forensic investigation reached its main decryption goal successfully.

The evidence preserved was checked for integrity and copied to a working location without changing the original evidence. The decompressed file was confirmed to be an IEEE 802.11 PCAP with 45,169 packets.

Aircrack-ng discovered the historical wireless network cgnetwork, BSSID 00:26:66:55:97:D6, with WEP protection, and 15,477 captured IVs. WEP40 key

A4:3D:F6:F3:74 was successfully recovered and aircrack-ng reported 100% correct decryption.

Then the key was validated with Airdecap-ng. The tool read all 45169 packets, found 15477 WEP data packets and decrypted all 15477 successfully having zero corrupted WEP packets.

The decrypted capture was stored separately as:

wep40_original_working-dec.pcap

The original wireless capture was saved as:

pcap/wep40_original_working.pcap

The investigation therefore demonstrates successful offline forensic recovery and validation of a WEP40 key from the provided historical capture and successful decryption of the associated WEP data.

If the lab marking rubric requires it, further post-decryption endpoint, conversation, and object-recovery analysis will be documented.